

Zeeshan Fazal, PhD

Bioinformatics Scientist · Clinical Genomics · Transcriptomics · Multi-Omics Integration

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818

TOTAL CITATIONS

14

H-INDEX

18+

PUBLICATIONS

8+

NIH RESEARCH PROGRAMS

9+

YEARS EXPERIENCE

CLINICAL GENOMICS

GATK / HaploTypeCaller Mutect2
GEMINI Exomiser VEP
ANNOVAR DeepVariant
DeepSomatic Panel of Normals
mtDNA / Heteroplasmy
Telomere Analysis IUIS Gene Lists

TRANSCRIPTOMICS

scRNA-seq Bulk RNA-seq
Seurat Cell Ranger
STAR / Salmon DESeq2 / edgeR
GSEA ChIP-Seq ATAC-Seq
EPIC Arrays

PIPELINES & INFRASTRUCTURE

Snakemake SLURM / HPC
Git / GitHub Docker / Conda
Cloud (AWS) Claude Code
GitHub Copilot

LANGUAGES

Python (Pandas, NumPy, sklearn)
R / Bioconductor Bash / Shell
SQL

DATABASES

gnomAD ClinVar / OMIM
TCGA / GEO cBioPortal
HPA / CCLE

EDUCATION

PhD, Animal Sciences

Bioinformatics Concentration
Univ. of Illinois (UIUC)
2017

MS, Bioinformatics

Univ. of Illinois (UIUC)
2012

AWARDS & HONORS

NASA Group Achievement Award, 2018

UIUC Graduate Fellowship, 2017

Government of Pakistan Merit Fellowship, 2010

Institute Bronze Medal, BS Bioinformatics, 2009

MEMBERSHIPS

American Society of Human Genetics (ASHG), 2024–
American Association for Cancer Research (AACR), 2019–
National Postdoctoral Association, 2019–

PROFESSIONAL SUMMARY

Bioinformatics PhD with 9+ years of experience in clinical genomics, transcriptomics, and multi-omics integration across immunology, rare disease, cancer, and infectious disease research — spanning NIH/NIAID, Fate Therapeutics, Stanford University, and UIUC. Deep expertise in WGS/WES somatic and germline variant calling (GATK, Mutect2), bulk and single-cell RNA-seq (including viral infection and immune response contexts), and scalable HPC pipeline development (Snakemake, SLURM). Consistent record of independently leading multi-cohort analyses and translating results into peer-reviewed publications (h-index 14, 818 citations, 18+ papers). Proficient in AI-assisted development with GitHub and Claude Code.

PROFESSIONAL EXPERIENCE

Computational Genomics Specialist, Clinical

Guidehouse Digital / NIH-NIAID (Contractor)

Feb 2023 – Present

- Somatic Variant Calling:** Built and deployed production-grade Mutect2/GATK workflows for large-scale WGS cohorts; constructed Panel of Normals (PONs) supporting ongoing immunology and molecular aging programs. Published: "Clinical relevance of mosaic variants detected by exome sequencing."
- Germline & Rare Disease Genomics:** Led end-to-end germline analysis (GATK, GEMINI, Exomiser) across 8+ rare disease, immunodeficiency, and infectious disease programs; integrated IUIS 2024 gene lists, inheritance modeling, and phenotype-driven filters for clinical-grade variant prioritization.
- Multi-Modal Variant Annotation:** Executed GEMINI workflows in three modes (gene panel, exome-level, genome-level) across large WGS/WES cohorts, delivering consistent, publication-ready outputs across multiple concurrent PI programs.
- Bulk RNA-seq:** Performed differential expression and pathway analyses of drug-exposed immune cells; characterized transcriptomic responses to antiretroviral therapies across multiple donors and treatment conditions.
- Single-Cell RNA-seq:** Applied scRNA-seq workflows to resolve immune cell heterogeneity and transcriptomic states in viral infection and inflammatory disease contexts, supporting host-pathogen research programs.
- Multi-Omics & Aging:** Integrated somatic mutation burden and telomere length estimation to define molecular aging biomarkers; developed mitochondrial variant calling with heteroplasmy estimation, adopted by the NIH central sequencing program.
- Infrastructure:** Built modular, reproducible pipelines (SLURM/Snakemake) with GitHub and Claude Code; enabled fast, reliable analytical turnaround across 8+ simultaneous research programs.

Bioinformatics Scientist

Fate Therapeutics, Inc., San Diego, CA

Apr 2022 – Feb 2023

- Developed transcriptomic-based phenotype classification signatures supporting CAR-T and NK cell immunotherapy discovery programs.
- Designed genomic alignment workflows for custom-engineered cell therapy clones, improving throughput and clone-level resolution.
- Integrated transcriptomic and genomic datasets for therapeutic target identification across oncology indications.

Bioinformatics Postdoctoral Researcher / Research Scientist

University of Illinois at Urbana-Champaign

Jun 2018 – Apr 2022

- Led integrative analysis of chemotherapy-resistant cancer models using RNA-Seq, ChIP-Seq, and Illumina EPIC methylation arrays; identified polycomb complex dysregulation as a key resistance mechanism.
- Provided bioinformatics support for a Phase 1 clinical trial combining epigenetic and chemotherapy agents; published in *Cancer Medicine*.
- Produced 8+ peer-reviewed publications; first-author papers in *Epigenetics* and *ACS Nano*.

Bioinformatics Postdoctoral Researcher

Stanford University, Stanford, CA

Sep 2017 – May 2018

- Performed high-throughput virtual screening of ~2 million compounds to identify novel inhibitors of bacterial antioxidant systems; part of NASA-funded microgravity antibiotic resistance research.
- Recipient of the **NASA Group Achievement Award** (2018) for contributions to the CubeSat biological experiment.

SELECTED PUBLICATIONS (18+ TOTAL · 818 CITATIONS · H-INDEX 14 · [FULL LIST ON GOOGLE SCHOLAR](#))

1. **Fazal Z**, et al. "Clinical relevance of mosaic variants detected by exome sequencing." *Translational and Clinical Immunology*, 2026.
2. Shokry D, ..., **Fazal Z**, et al. "Refractory testicular germ cell tumors are highly sensitive to the targeting of polycomb pathway demethylases KDM6A and KDM6B." *Cell Communication and Signaling*, 22(1), 528, 2024.
3. Boyd RI, ..., **Fazal Z**, et al. "Perfluorooctanesulfonic Acid Alters Pro-Cancer Phenotypes and Metabolic and Transcriptional Signatures in Testicular Germ Cell Tumors." *Toxics*, 12(4), 232, 2024.
4. Boyd RI, Ahmad S, Singh R, **Fazal Z**, et al. "Toward a mechanistic understanding of poly- and perfluoroalkylated substances and cancer." *Cancers*, 14(12), 2919, 2022. [\[131 citations\]](#)
5. Singh R, **Fazal Z**, et al. "Reciprocal epigenetic remodeling controls testicular cancer hypersensitivity to hypomethylating agents and chemotherapy." *Molecular Oncology*, 16(3), 683–698, 2022.
6. **Fazal Z**, Singh R, Fang F, et al. "Hypermethylation and global remodelling of DNA methylation is associated with acquired cisplatin resistance in testicular germ cell tumours." *Epigenetics*, 16(10), 1071–1084, 2021.
7. Albany C, **Fazal Z**, Singh R, et al. "A phase 1 study of combined guadecitabine and cisplatin in platinum refractory germ cell cancer." *Cancer Medicine*, 2020.
8. **Fazal Z**, Pelowitz J, et al. "Three-Dimensional Encapsulation of *Saccharomyces cerevisiae* in Silicate Matrices Creates Distinct Metabolic States as Revealed by Gene Chip Analysis." *ACS Nano*, 11(4), 3560–3575, 2017.
9. Jakobsson E, ..., **Fazal Z**, et al. "Towards a unified understanding of lithium action in basic biology and its significance for applied biology." *Journal of Membrane Biology*, 250, 587–604, 2017. [\[201 citations\]](#)
10. Akhtar MN, Bukhari SA, **Fazal Z**, Qamar R, Shahmuradov IA. "POLYAR, a new computer program for prediction of poly(A) sites in human sequences." *BMC Genomics*, 11, 1–10, 2010. [\[64 citations\]](#)